



Semester I Examinations 2017/2018

Exam Code(s) 1SD3, 1SD1, 1MF1
Exam(s) Higher Diploma in Software Design and Development
MSc in Software Design and Development

Module Code(s) CT537

Module(s) Software Engineering Methods

Paper No. 1
Repeat Paper

External Examiner(s) Prof. Pier Luca Lanzi
Internal Examiner(s) Dr. Michael Schukat
Ms. Karen Young*

Instructions: Candidates should answer **Question 1** and any **two other** questions.
All questions carry equal marks.

Duration 2 hours

No. of Pages 5

Requirements:

Release in Exam Venue Yes ☒ No ☐

MCQ Yes ☐ No ☒

Handout None

Statistical/ Log Tables None

Cambridge Tables None

Graph Paper None

Log Graph Paper None

Other Materials None

Graphic material in colour Yes ☐ No ☒

- 1.(a) Draw a context diagram, a systems level DFD and one third level DFD**
to represent the data and processing involved in the following system description of the *Lodge* hotel:

The *Lodge* Hotel is a small country hotel that has recently been taken over by a larger hotel company, the *Excel* Company. You are a systems analyst who has been employed by *Excel* to review current operations at the *Lodge* Hotel with a view to upgrading and expanding its current computer system. At present the *Lodge* Hotel has a small computer-based system that handles its operations. The system supports the reservation of rooms by phone, e-mail, or in person. The hotel has just begun work on a Website that will allow online bookings to be processed. Rooms are charged at an overnight rate, with special offers and seasonal reductions possible, and reductions are applied after 3 consecutive nights stay. Breakfast is charged separately. Bills are calculated based on accommodation, mini-bar sales, breakfast, and other charges (e.g. phone, laundry). Guests who pre-book rooms are required to pay a minimum deposit at booking time. Guests may pay for the rooms in full in advance. At the end of their stay, guests settle their remainder of account and are issued with a receipt. After eight stays at the *Lodge* hotel, guests are rewarded with a loyalty discount scheme of 10% reduction on accommodation rates.

The hotel accepts credit card payments, provides EFTPOS facilities, and cash or cheque. Each week a room occupancy report and a financial report are forwarded to the *Excel* Company.

(15)

- (b)** Assume you are leading the project to develop the hotel system described above. Provide a brief report detailing how the **implementation** and **testing** of the system will be approached. The report should detail the design method, programming language and framework employed, stating all assumptions made.

(5)

2. (a) Prepare an *Activity* diagram for the *self service petrol* system described below:

You have been asked to create a system to automate transactions at a self-service petrol station. The petrol station has only three pumps: one for diesel, one for unleaded, and one for leaded. The station also sells food and other items. All of these items have a barcode. A separate database has been populated with the barcode, description and price of each item sold.

The cashier obtains from the customer the number of the petrol pump and enters that information into the system. The system must then interact with the computer system on the appropriate pump to get the amount of fuel purchased in litres and the cost in pounds. The cashier then scans the barcodes of the customers other purchases. The system interacts with this database to get the description and price of items scanned by the cashier. A total amount due is computed by adding together the fuel purchase and the other purchases.

If the sale is paid with cash, the customer pays the cashier and receives the food purchased, a receipt, and any change due.

If the sale is paid with a credit card, the cashier scans the card using a special scanner to read the account number. The system must then make a call via modem to a central credit card agency computer to verify that the account is OK and that the balance is sufficient to cover the transaction. If the card is OK and the balance is sufficient, the system tells the bank to debit the total from the customer's account. The system must print a receipt listing the items purchased, the total amount due, and the customer's credit card number. The customer is given the food purchased and a customer copy of the credit receipt. The original credit receipt is forwarded by post to the bank that handles all credit card transactions. If the card is not OK or the balance is insufficient the transaction is cancelled completely.

(8)

(b) Elaborate on the diverse contributions of an **Activity** diagram during the full OO development lifecycle, illustrating your response with examples as appropriate.

(2)

(c) Compare and contrast the Structured Analysis **DFD** and the UML's **Activity Diagram** as two examples of process modelling techniques.

(4)

(d) Discuss the advantages and disadvantages of the approach taken by a company you work for in the following situation:

The company you work for develops internet applications. To reduce time to market, the company has decided to dispense altogether with integration testing. Instead, the company will rely on beta testing, in which free trial versions of all new software will be sent to some existing customers to try out, with the agreement that they will report any problems they encounter.

(6)

3. (a) In software development projects, deciding “**what**” to say is recognised as being more difficult than “**how**” to say it. Recent trends have further complicated this process for developers. Discuss what **requirements processes and techniques** you would use as a project manager to handle the following requirements challenges, illustrating your response with appropriate examples:

- (i) continuously changing user requirements
- (ii) changing technology deployment environments

(8)

(b) Cost estimates are inherently risky, irrespective of the estimation technique used. How would you reduce the risks inherent in a cost estimate?

(2)

(c) Suppose you are managing a project which is getting behind schedule. Possible actions include renegotiating the time schedule, adding people to the project, and renegotiating quality requirements. In which ways can these actions shorten the time schedule? What other actions could you take to finish the project on time?

(5)

(d) Software Design is both a process and a product. What is a software design? How does a software design differ from a software program? How is the quality of a software design assessed?

(5)

4. (a) Prepare a **Use Case diagram** for the *Health club* system described below and expand **one** of the identified use cases to a full textual description.

When members join the Fit4life health club, they pay a fee for a certain length of time. Most memberships are for one year, but memberships as short as two months are available. Throughout the year, the health club offers a variety of discounts on their regular membership prices (e.g., two memberships for the price of one for Valentine's Day). It is common for members to pay different amounts for the same length of membership. The club wants to mail out reminder letters to members asking them to renew their memberships one month before their memberships expire. One of the problems in the health club industry is the high turnover rate of members. While some members remain active for many years, about half of the members do not renew their memberships. This is a major problem, because the health club spends a lot in advertising to attract each new member. The manager wants the system to track each time a member comes into the club. The system will then identify the heavy users, and generate a report so the manager can ask them to renew their memberships early, perhaps offering them a reduced rate for early renewal. Likewise, the system should identify members who have not visited the club in more than a month, so the manager can call them and attempt to re-interest them in the club.

(8)

(b) You are the team lead of a systems design team that is implementing a multiple module project with an expected duration of two years. Your client just sent you the memo included below. Write a memo back to your client answering their questions.

Dear Lead,

*I was reading **IT Update** magazine and an article mentioned that many project teams now use prototyping and agile techniques to speed up their systems development processes. What are these techniques? Will they save us time and money? What problems might we encounter if we use them?*

*Regards,
XYZ.*

(6)

(c) How would you define **software quality**? Evaluate the contribution of the software engineering techniques covered in this course to improving the quality of software produced.

(6)